

Protein Supply Trends in Bangladesh

The case study examines how protein supply in Bangladesh has changed over time and how different food groups contribute to protein availability.

Six selected food groups are:

- Fish
- Meat
- Milk
- Eggs
- Pulses
- Cereals

Main Data Variable:

Primary measure: Protein supply quantity (g/capita/day). It tells us approximately how many grams of protein are available per person per day from each food category.

Key Calculations:

A. Protein supply trend

For each food group:

$$\text{Change} = (\text{Protein}_{2023} - \text{Protein}_{2010})$$

$$\text{Percentage change} = ((\text{Protein}_{2023} - \text{Protein}_{2010}) * 100) / \text{Protein}_{2010}$$

B. Contribution/share

For the six selected food groups:

$$\text{Food Group Share} = (\text{Protein from Food Group} * 100) / \text{Protein from Six Selected Groups}$$

For example:

$$\text{Fish Share} = (\text{Fish} * 100) / (\text{Fish} + \text{Meat} + \text{Milk} + \text{Eggs} + \text{Pulses} + \text{Cereals})$$

C. Animal vs plant

Animal Share = ((Fish+Meat+Milk+Eggs)*100)/Six Selected Groups

Plant Share= ((Pulses+Cereals)*100)/ Six Selected Groups

Interpretation

Between 2010 and 2023, protein supply from most of the six selected food groups increased. The largest increase occurred in milk, which rose by 247.5%, followed by eggs (156.5%), pulses (75.8%), and fish (40.1%).

Meat increased by only 15.2%, while cereal-derived protein declined slightly by 3.0%. Despite this decline, cereals remained the dominant source of protein, accounting for 73.8% of the protein supplied by the six food groups in 2010 and 60.1% in 2023. The decline in cereal share was accompanied by increasing contributions from milk, pulses, fish, and eggs.

Consequently, the share of protein from animal-source foods increased from 19.8% in 2010 to 30.4% in 2023, while the plant-source share declined from 80.2% to 69.6%. Overall, the results indicate a shift in the composition of protein supply away from heavy dependence on cereals toward a more diversified contribution from animal-source foods and pulses.